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Mini Project Report on
Cloud-Based Event Management Portal using Microsoft Azure
Submitted in partial fulfillment of the requirements for the award of the degree

Master of Technology

in

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CHAPTER 1

INTRODUCTION

1.1 Background

The rapid growth of cloud computing has transformed the way web applications are developed, deployed, and maintained. Organizations and educational institutions increasingly rely on cloud platforms to host applications because they provide high availability, scalability, reliability, and secure access from any location. Cloud platforms eliminate the need for maintaining expensive local servers while offering automatic deployment and continuous integration services.

Event management is an essential activity in colleges, universities, companies, and organizations. Traditional methods of managing events often involve manual registration forms, spreadsheets, and paper-based records, making the process time-consuming and prone to errors. Participants may face difficulties in obtaining event information, registering for events, and receiving timely updates.

The **Cloud-Based Event Management Portal** is designed to simplify the entire event management process by providing a centralized web application hosted on Microsoft Azure. The application allows users to browse upcoming events, view event schedules, register online, and receive registration confirmation through an intuitive interface.

The project is developed using **HTML, CSS, and JavaScript** in Visual Studio Code, with the source code managed through **GitHub** and deployed using **Azure DevOps Pipelines**. The integration of GitHub with Azure enables Continuous Integration and Continuous Deployment (CI/CD), ensuring that every change made to the application is automatically deployed to the cloud. This approach demonstrates the practical implementation of cloud computing technologies in modern web application development while providing a reliable, scalable, and easily accessible event management solution.

1.2 Problem Statement

Many educational institutions and organizations continue to manage events using manual registration methods or basic web pages that lack automation and cloud deployment. These traditional systems make it difficult to maintain event information, manage participant registrations, and update content efficiently. Manual processes increase administrative workload and are susceptible to errors, duplicate registrations, and delayed communication.

Additionally, locally hosted applications require continuous maintenance and do not provide the flexibility of remote accessibility or automated deployment. Every update requires manual intervention, increasing development time and reducing system reliability.

To address these limitations, this project proposes a Cloud-Based Event Management Portal deployed on Microsoft Azure. The application uses Azure cloud services and GitHub integration to provide automated deployment, centralized management, secure hosting, and easy accessible.

1.3 Objectives

- To develop a cloud-based Event Management Portal using Microsoft Azure.
- To provide a centralized platform for displaying event information.
- To allow users to register for events through an online registration form.
- To display upcoming events and event schedules in an organized manner.
- To implement a responsive and user-friendly web interface.
- To integrate GitHub with Azure DevOps for Continuous Integration and Continuous Deployment (CI/CD).
- To reduce manual effort in event registration and information management.
- To demonstrate the practical implementation of cloud computing concepts.
- To provide secure, reliable, and scalable web application hosting using Microsoft Azure.
- To enhance user accessibility by making the application available online.

1.4 Scope of the Project

The scope of the proposed project is to develop a cloud-based web application that provides an efficient platform for managing event-related information. The application enables users to view upcoming events, browse event schedules, register for events, and contact organizers through an interactive web interface.

The project primarily focuses on cloud deployment using Microsoft Azure, version control through GitHub, and automated deployment using Azure DevOps Pipelines. The portal is designed to be responsive, allowing users to access the application from desktops, laptops, tablets, and mobile devices.

Although the current implementation focuses on static event information and registration functionality, the application can be extended in the future by integrating cloud databases, user authentication, email notifications, payment gateways, QR code-based event passes, and real-time event management features.

1.5 Research Gap

Many existing event management websites are either desktop-based or hosted on traditional web servers that require manual deployment and maintenance. Such systems often lack automated deployment mechanisms, cloud scalability, and centralized application management. Updating application content frequently requires technical intervention, making the maintenance process inefficient.

Several commercial event management platforms provide advanced functionalities; however, they are often expensive and contain features that may not be necessary for small educational institutions and student projects. Additionally, many academic projects focus only on website development without demonstrating cloud deployment or DevOps practices.

This project addresses these limitations by developing a cloud-based Event Management Portal using Microsoft Azure, GitHub, and Azure DevOps Pipelines. The proposed solution demonstrates cloud application deployment, version control, continuous integration, and continuous deployment while maintaining a simple and user-friendly interface suitable for educational purposes.

1.6 Proposed System

The proposed system is a cloud-hosted Event Management Portal developed using HTML, CSS, and JavaScript. The application provides users with a centralized platform to access event information, browse schedules, and register for various events through an intuitive web interface.

The application source code is maintained using GitHub, while Azure DevOps Pipelines automate the deployment process whenever changes are committed to the repository. Microsoft Azure hosts the application, ensuring high availability, scalability, and reliable access over the internet.

The portal consists of multiple modules, including Home, About, Events, Event Registration, Event Schedule, Gallery, and Contact pages. The responsive user interface enables users to access the application from different devices without additional software installation.

Compared to traditional static websites, the proposed solution demonstrates modern cloud application deployment practices using Continuous Integration and Continuous Deployment (CI/CD). The application provides better maintainability, simplified deployment, improved accessibility, and reliable cloud hosting through Microsoft Azure.

1.7 Contributions

The major contributions of the proposed project are:

- Developed a cloud-based Event Management Portal using Microsoft Azure.
- Designed a responsive and interactive user interface using HTML, CSS, and JavaScript.
- Integrated GitHub for version control and collaborative development.
- Configured Azure DevOps Pipelines for Continuous Integration and Continuous Deployment (CI/CD).
- Automated cloud deployment whenever source code is updated.
- Reduced manual deployment effort through Azure cloud services.
- Improved accessibility by hosting the application on Microsoft Azure.

- Demonstrated practical implementation of cloud computing concepts in web application deployment.
- Created a scalable and maintainable application architecture suitable for future enhancements.
- Provided an educational example of integrating GitHub, Azure DevOps, and Microsoft Azure into a complete cloud-based web application.

1.8 Organization of the Report

The remainder of this report is organized as follows:

- **Chapter 2** presents the literature survey on cloud storage systems, Microsoft Azure Blob Storage, secure authentication, and related cloud-based file management techniques.
- **Chapter 3** discusses the system analysis and design, including the existing system, proposed system, system architecture, flow diagram, and use case diagram.
- **Chapter 4** explains the methodology adopted for user authentication, file management, Azure Blob Storage integration, and database operations.
- **Chapter 5** describes the implementation details, technologies used, software tools, and project modules.
- **Chapter 6** presents the application results, screenshots, and discussion of the developed CloudVault system.
- **Chapter 7** concludes the report by summarizing the project outcomes and suggesting possible future enhancements.

CHAPTER 2

LITERATURE SURVEY

2.1 Introduction

Cloud computing has significantly transformed the development and deployment of modern web applications by providing scalable, reliable, and cost-effective infrastructure. Organizations and educational institutions increasingly use cloud platforms such as Microsoft Azure to host applications, ensuring high availability, secure access, and simplified maintenance. Along with cloud hosting, version control systems like GitHub and DevOps practices have become essential for collaborative software development and automated deployment.

Event management systems are widely used to organize conferences, workshops, seminars, technical fests, and cultural programs. Traditional event management methods often rely on manual registration and static web pages, which are difficult to maintain and update. Cloud-based event management portals address these challenges by offering centralized access to event information, online registration, and automated deployment through cloud platforms.

This chapter reviews existing research related to cloud computing, Microsoft Azure, GitHub-based version control, and web-based event management systems. It discusses their methodologies, advantages, limitations, and identifies the research gap that motivated the development of the proposed **Cloud-Based Event Management Portal using Microsoft Azure and GitHub**.

2.2 Literature Review

Paper 1

Microsoft Azure App Service: A Scalable Platform for Cloud Web Applications (2023)

Authors: Microsoft Azure Documentation

This work describes Microsoft Azure App Service as a cloud platform that enables developers to build, deploy, and host web applications without managing physical servers. It supports automatic scaling, secure deployment, high availability, and seamless integration with Azure DevOps and

GitHub. Azure App Service simplifies application deployment while improving performance and reliability.

Advantages

- Provides secure and reliable cloud hosting.
- Supports automatic deployment and scaling.
- Easily integrates with GitHub and Azure DevOps.

Limitations

- Requires internet connectivity.
- Advanced cloud features may require paid subscription plans.

Initial Azure configuration may be challenging for beginners.

Paper 2

GitHub and CI/CD for Modern Web Application Development (2022)

Authors: GitHub Documentation

This study explains the importance of GitHub as a distributed version control platform that enables collaborative software development. By integrating GitHub with Continuous Integration and Continuous Deployment (CI/CD) pipelines, developers can automatically build, test, and deploy web applications whenever code changes are committed. This approach improves software quality and reduces manual deployment effort.

Advantages

- Efficient version control and collaboration.
- Automated application deployment.
- Simplifies project maintenance.

Limitations

- Requires understanding of Git commands.
- Pipeline configuration can be complex for beginners.
- Continuous internet connectivity is required.

Paper 3

Web-Based Event Management Systems for Educational Institutions (2024)

Authors: Various Researchers

This study focuses on web-based event management systems that simplify event organization through online registration and centralized event information. The research highlights that digital event management improves user experience, reduces administrative workload, and enables participants to access event schedules and registration details from any location.

Advantages

- Simplifies event registration.
- Improves accessibility for participants.
- Reduces manual paperwork and administrative effort.

Limitations

- Requires regular updates to event information.
- Performance depends on server availability.
- Basic systems often lack cloud deployment and automation.

2.3 Comparative Analysis

Paper	Technique	Advantages	Limitations
Microsoft Azure App Service	Cloud Web Hosting	Scalable, secure, and reliable application hosting	Requires internet connectivity
GitHub with CI/CD	Version Control & Automated Deployment	Efficient collaboration and automatic deployment	Requires Git and pipeline configuration
Web-Based Event Management System	Online Event Management	Easy registration and centralized event information	Requires continuous maintenance and updates

2.4 Research Gap

The literature survey indicates that existing event management systems primarily focus on online registration and basic website functionality, while cloud hosting platforms mainly emphasize scalable application deployment. Similarly, GitHub provides efficient version control and collaboration, but many educational projects do not demonstrate the complete integration of GitHub, Azure DevOps, and Microsoft Azure within a single application.

The major research gaps identified are:

- Limited educational projects demonstrating Microsoft Azure cloud deployment.
- Lack of integration between GitHub, Azure DevOps, and web-based event management systems.
- Minimal emphasis on Continuous Integration and Continuous Deployment (CI/CD).
- Limited demonstration of cloud-based application deployment for academic mini projects.
- Few simple and scalable event management applications developed specifically for educational purposes.

The proposed **Cloud-Based Event Management Portal using Microsoft Azure and GitHub** addresses these limitations by integrating a responsive web application with GitHub version control, Azure DevOps CI/CD pipelines, and Microsoft Azure cloud hosting. The project demonstrates practical cloud computing concepts while providing a simple, user-friendly, and scalable event management solution suitable for educational institutions.

CHAPTER 3

SYSTEM ANALYSIS AND DESIGN

3.1 Existing System

Traditional event management systems are primarily based on manual registration methods, spreadsheets, or simple static websites. Event organizers often collect participant information using paper forms or online forms that require manual processing. These methods make it difficult to maintain event records, update schedules, and communicate changes efficiently.

Several commercial event management platforms provide online registration and scheduling features. However, these systems are subscription-based and do not demonstrate the implementation of cloud deployment, version control, or Continuous Integration and Continuous Deployment (CI/CD), making them less suitable for educational learning and mini projects.

Disadvantages of Existing System

- Manual registration and event management.
- Time-consuming process for updating event details.
- Limited accessibility from different locations.
- No automated deployment mechanism.
- Lack of cloud hosting and scalability.
- Difficult to maintain multiple versions of the application.

3.2 Proposed System

The proposed **Cloud-Based Event Management Portal** is a responsive web application developed using HTML, CSS, and JavaScript. It enables users to browse upcoming events, view schedules, and register online through an interactive interface. The application is hosted on Microsoft Azure, providing reliable cloud access from any internet-enabled device.

The source code is maintained using GitHub, and Azure DevOps Pipelines automate the build and deployment process. Whenever changes are pushed to the GitHub repository, the Azure

3.3 System Architecture

The proposed Event Management Portal follows a cloud-based client-server architecture integrated with GitHub and Microsoft Azure services.

- i. The user accesses the Event Management Portal through a web browser.
- ii. The frontend developed using HTML, CSS, and JavaScript displays event information, schedules, and registration forms.
- iii. JavaScript validates user inputs and manages interactions such as form submission and confirmation messages.
- iv. The application source code is maintained in a GitHub repository, allowing efficient version control and collaboration.
- v. Azure DevOps Pipelines automatically detect changes in the GitHub repository, build the application, and deploy the latest version to Microsoft Azure App Service.
- vi. Microsoft Azure hosts the application securely, allowing users to access the updated Event Management Portal from anywhere through the internet.
- vii. Azure DevOps Pipeline
- viii. Azure DevOps Pipelines monitor the GitHub repository for code changes. Whenever updates are pushed, the pipeline automatically builds and deploys the latest version of the application.
- ix. Microsoft Azure App Service
- x. Microsoft Azure App Service hosts the Event Management Portal, providing secure, reliable, and scalable cloud hosting with high availability.
- xi. Cloud Deployment
- xii. The CI/CD process ensures that every code modification is automatically deployed to Azure without manual intervention, reducing deployment time and improving consistency.
- xiii. User Accessibility
- xiv. The deployed application can be accessed from desktops, laptops, tablets, and smartphones through any modern web browser with an internet connection.
- xv. System Maintenance

- xvi. Developers can update the application, add new events, or modify the website by making changes to the source code and pushing them to GitHub, where Azure DevOps automatically deploys the updated version.

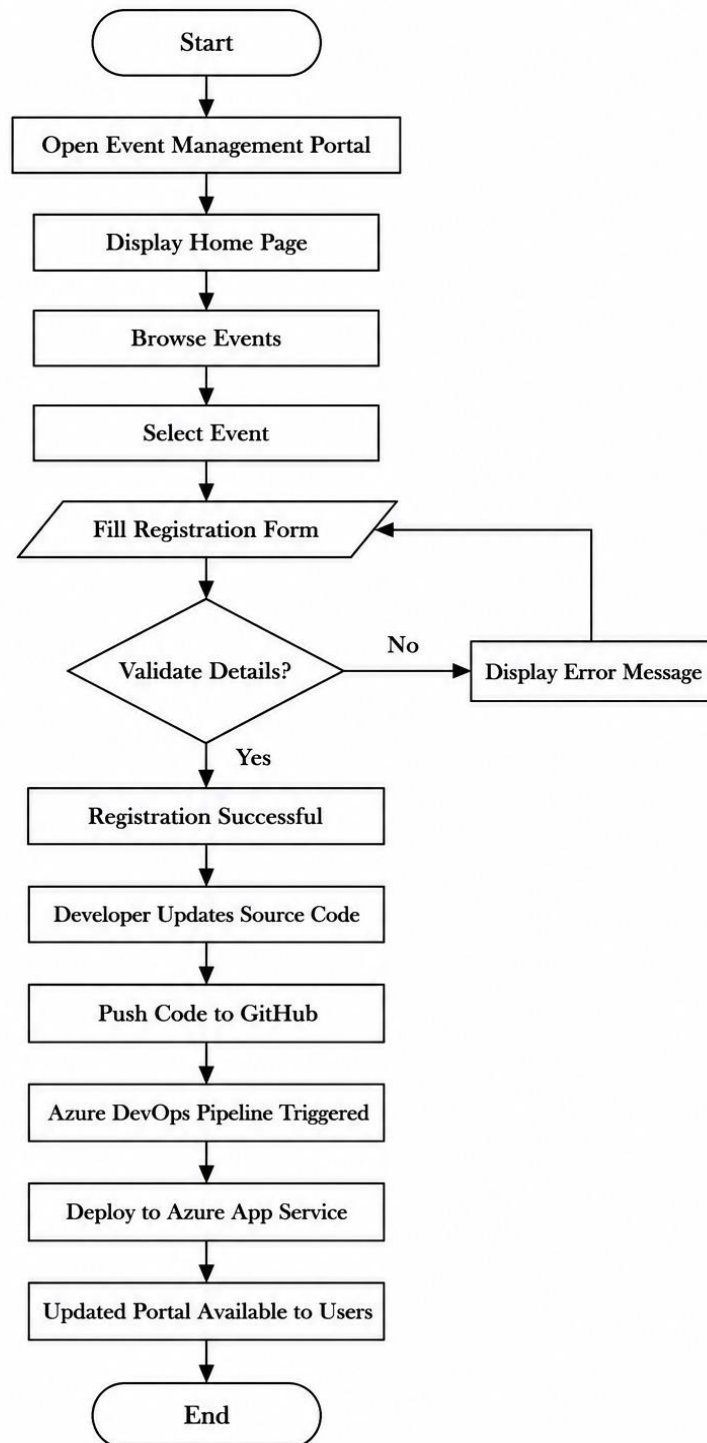
The proposed **Cloud-Based Event Management Portal** follows a cloud-based client-server architecture integrated with GitHub and Microsoft Azure services. The architecture consists of the presentation layer, application layer, version control, deployment pipeline, and cloud hosting platform, ensuring efficient application development and deployment. The Use Case Diagram illustrates the interaction between the **User**, **Developer**, and the **Cloud-Based Event Management Portal**. The user accesses the portal to browse events, view schedules, register for events, and contact organizers. The developer manages the application by updating the source code, pushing changes to GitHub, and deploying the application using Azure DevOps and Microsoft Azure.

Pipeline automatically builds and deploys the updated application to Azure App Service. This CI/CD workflow simplifies application maintenance, reduces manual deployment effort, and ensures that users always access the latest version of the portal.

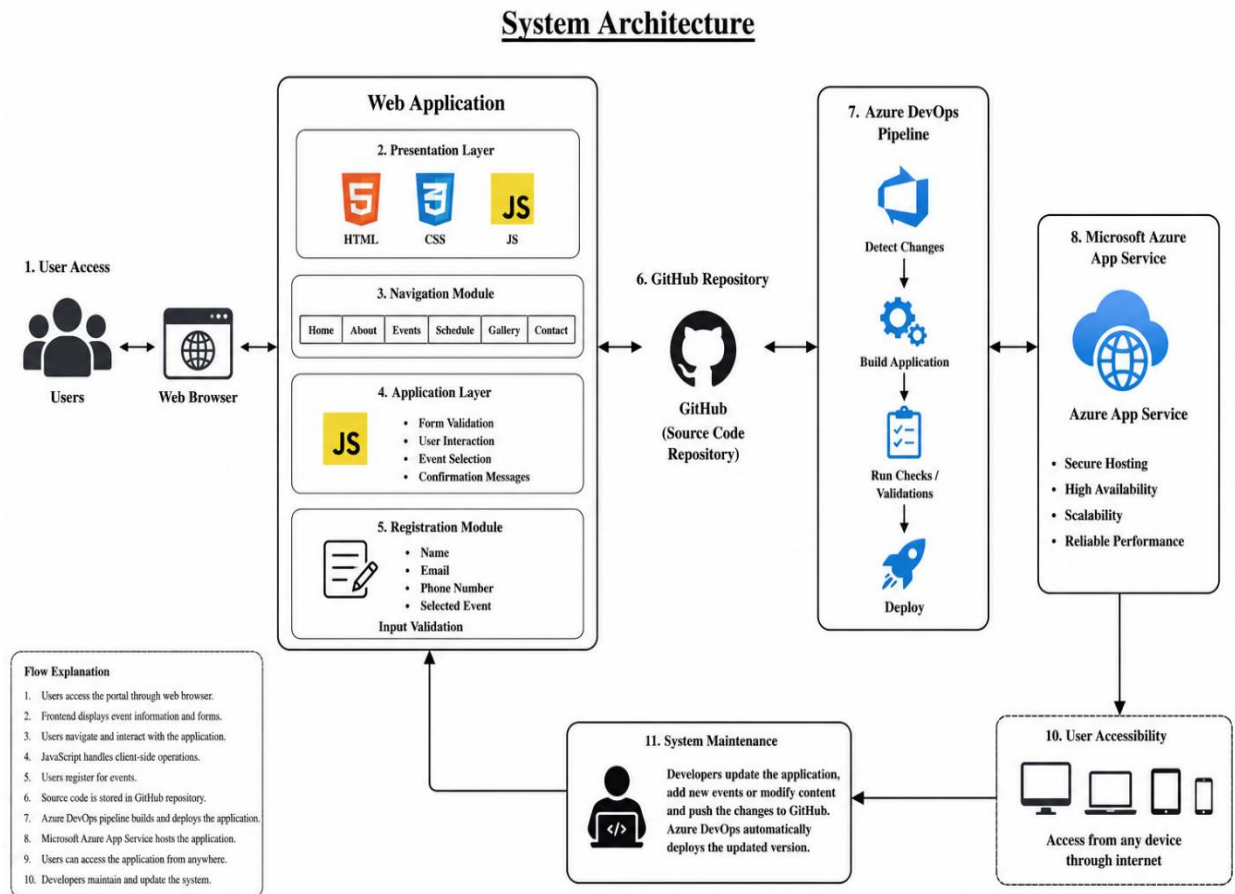
Advantages of Proposed System

- Responsive and user-friendly web interface.
- Cloud-based deployment using Microsoft Azure.
- Automated deployment through Azure DevOps Pipelines.
- Version control using GitHub.
- Easy event registration process.
- Scalable and maintainable application architecture.

3.4 Flow Diagram



3.5 Use Case Diagram



CHAPTER 4

METHODOLOGY

The proposed **Cloud-Based Event Management Portal using Microsoft Azure and GitHub** follows a structured methodology to provide an efficient and user-friendly platform for managing event information and registrations through a cloud-hosted web application. The development process begins with designing the frontend using **HTML, CSS, and JavaScript**, which provides an interactive and responsive interface for users. The application includes modules such as Home, About, Upcoming Events, Event Schedule, Event Registration, Gallery, and Contact. The responsive design ensures that the portal can be accessed seamlessly from desktops, laptops, tablets, and mobile devices.

Once the user accesses the application, the Home page displays information about upcoming events along with navigation options to different modules. Users can browse event details, schedules, venues, and other relevant information before selecting an event for registration. The registration module collects participant details such as name, email address, phone number, college name, and selected event. JavaScript performs client-side validation to verify that all mandatory fields are correctly entered before the registration request is processed. After successful validation, the application displays a confirmation message indicating that the registration has been completed successfully, thereby improving the overall user experience.

The complete application source code is maintained using **GitHub**, which provides version control and collaborative development. Developers create, modify, and manage project files in **Visual Studio Code** before committing the changes to the GitHub repository. GitHub maintains the project history, allowing developers to track code modifications, restore previous versions, and collaborate efficiently. This approach ensures organized source code management and simplifies future enhancements to the application.

The deployment process is automated using **Azure DevOps Pipelines**, which implement **Continuous Integration and Continuous Deployment (CI/CD)**. Whenever changes are pushed to the GitHub repository, Azure DevOps automatically detects the updates, builds the applicatio

CHAPTER 5

IMPLEMENTATION

The implementation of the **Cloud-Based Event Management Portal using Microsoft Azure and GitHub** was carried out using HTML, CSS, and JavaScript, following a modular web application architecture. The frontend was developed using HTML for structuring web pages, CSS for designing an attractive and responsive user interface, and JavaScript for implementing interactive functionalities such as form validation and user interactions. The application was developed and tested using **Visual Studio Code** on a Windows operating system.

The application consists of several modules, including the Home page, About page, Upcoming Events, Event Schedule, Event Registration, Gallery, and Contact page. Users can browse event details, view schedules, and register for events through an interactive registration form. JavaScript validates the user inputs to ensure that mandatory fields such as name, email address, phone number, and selected event are correctly entered before displaying a registration confirmation message. The responsive design enables users to access the application efficiently from desktops, laptops, tablets, and smartphones.

The complete source code of the application is maintained using **GitHub**, which provides version control and supports collaborative development. All project files are committed and pushed to the GitHub repository whenever modifications are made. **Azure DevOps Pipelines** are configured to automatically detect these changes and perform **Continuous Integration and Continuous Deployment (CI/CD)**. The pipeline builds the application, validates the project files, and deploys the latest version to **Microsoft Azure App Service**, eliminating the need for manual deployment and ensuring that the live application always reflects the most recent updates.

The completed Event Management Portal successfully integrates a responsive web interface, GitHub version control, Azure DevOps automation, and Microsoft Azure cloud hosting into a single cloud-based application. The modular implementation improves maintainability, scalability, and ease of future enhancements. Additional features such as user authentication, online payment integration, email notifications, cloud databases, QR-code-based event passes, and administrator

CHAPTER 6

RESULTS AND DISCUSSION

The proposed **CloudVault – Secure Cloud File Manager using Microsoft Azure** was successfully designed, developed, and tested. The application provides a secure and user-friendly platform for managing files in the cloud using Microsoft Azure Blob Storage. The integration of Flask, SQLite, and Azure Blob Storage enables users to perform essential file operations such as uploading, viewing, downloading, and deleting files through an intuitive web interface. User authentication ensures that only authorized users can access their personal files, thereby enhancing the overall security of the application.

The developed CloudVault application successfully achieved all the objectives defined for the project. It demonstrated secure user authentication, efficient cloud-based file management, and seamless integration with Microsoft Azure Blob Storage.

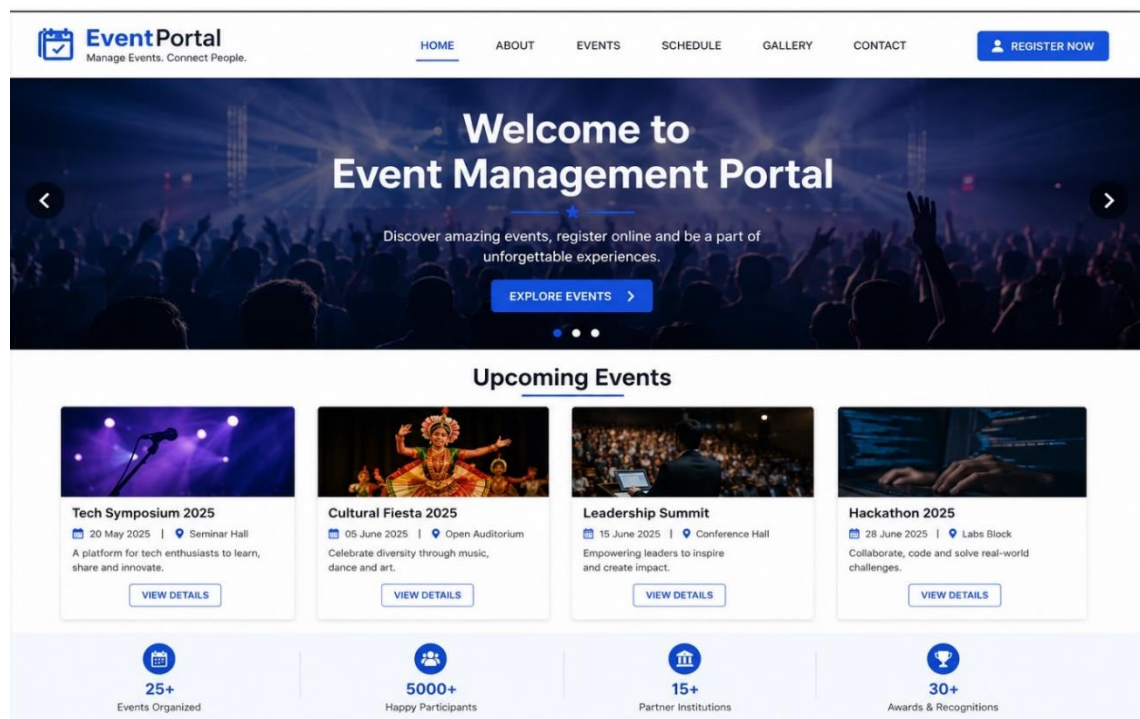


Fig 5.1 Home page of Event Management Portal

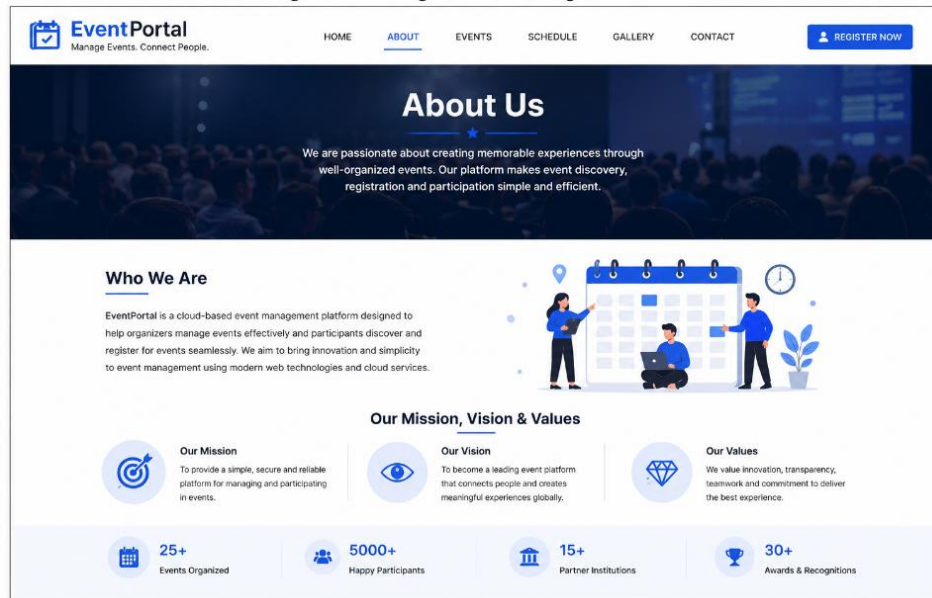


Fig 5.2 About Page of Event Management Portal

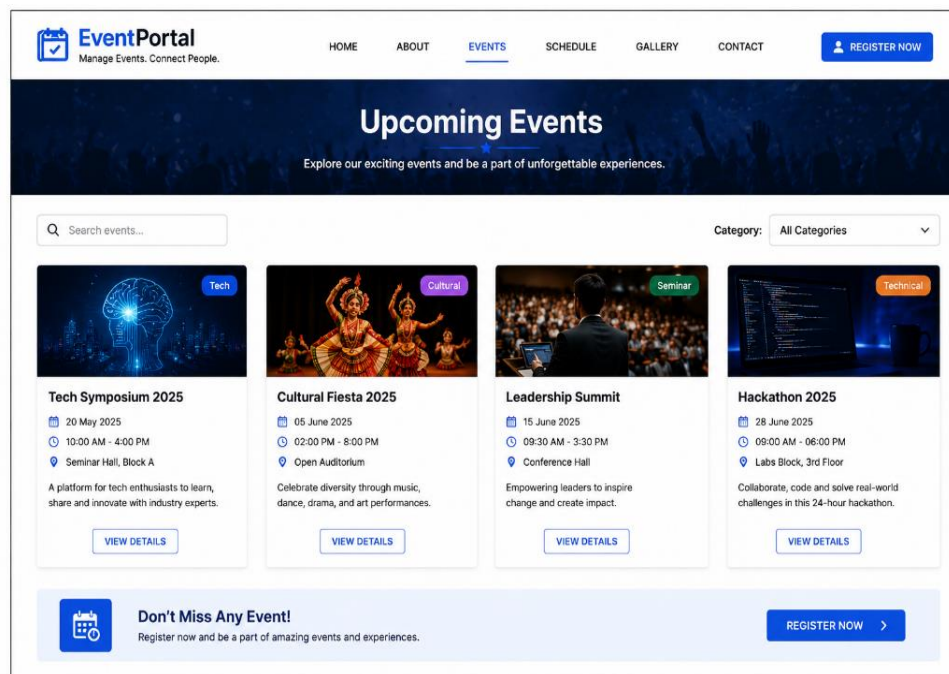
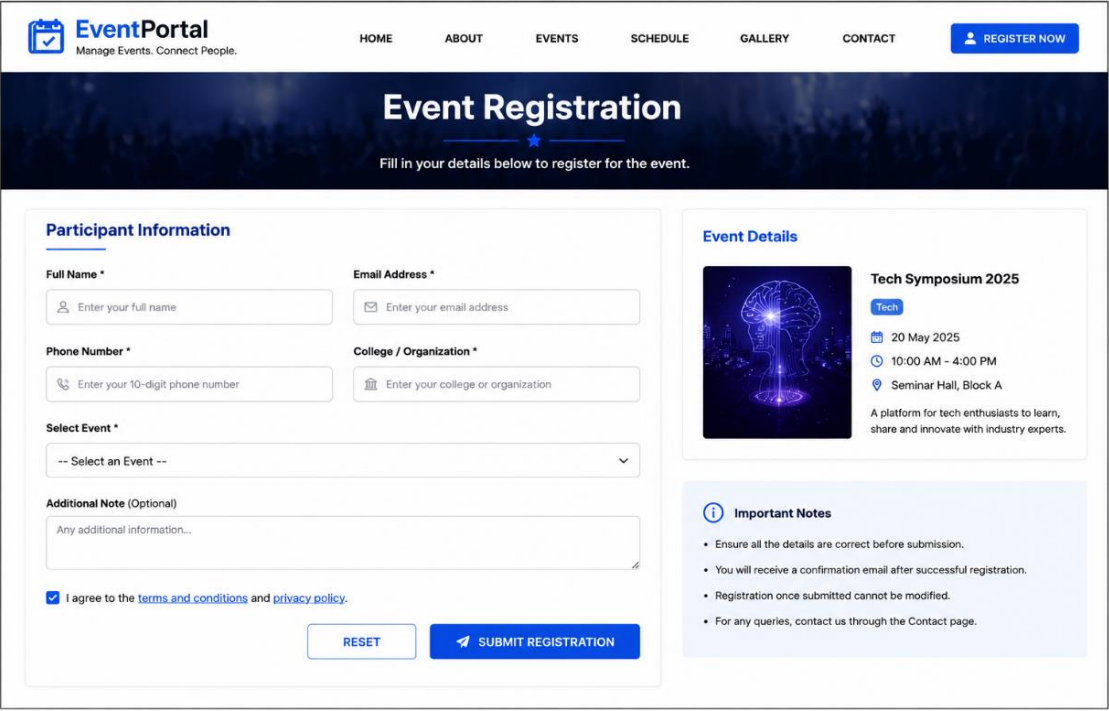


Fig 5.3 Upcoming Events Page of Event Management Portal



The screenshot shows the 'Event Registration' page of the EventPortal. The header includes the logo, navigation links (HOME, ABOUT, EVENTS, SCHEDULE, GALLERY, CONTACT), and a 'REGISTER NOW' button. The main heading is 'Event Registration' with a subtext 'Fill in your details below to register for the event.' The page is divided into two main sections: 'Participant Information' and 'Event Details'.

Participant Information

Full Name *
Enter your full name

Email Address *
Enter your email address

Phone Number *
Enter your 10-digit phone number

College / Organization *
Enter your college or organization

Select Event *
-- Select an Event --

Additional Note (Optional)
Any additional information...

☒ I agree to the [terms and conditions](#) and [privacy policy](#).

RESET SUBMIT REGISTRATION

Event Details

Tech Symposium 2025

Tech

20 May 2025

10:00 AM - 4:00 PM

Seminar Hall, Block A

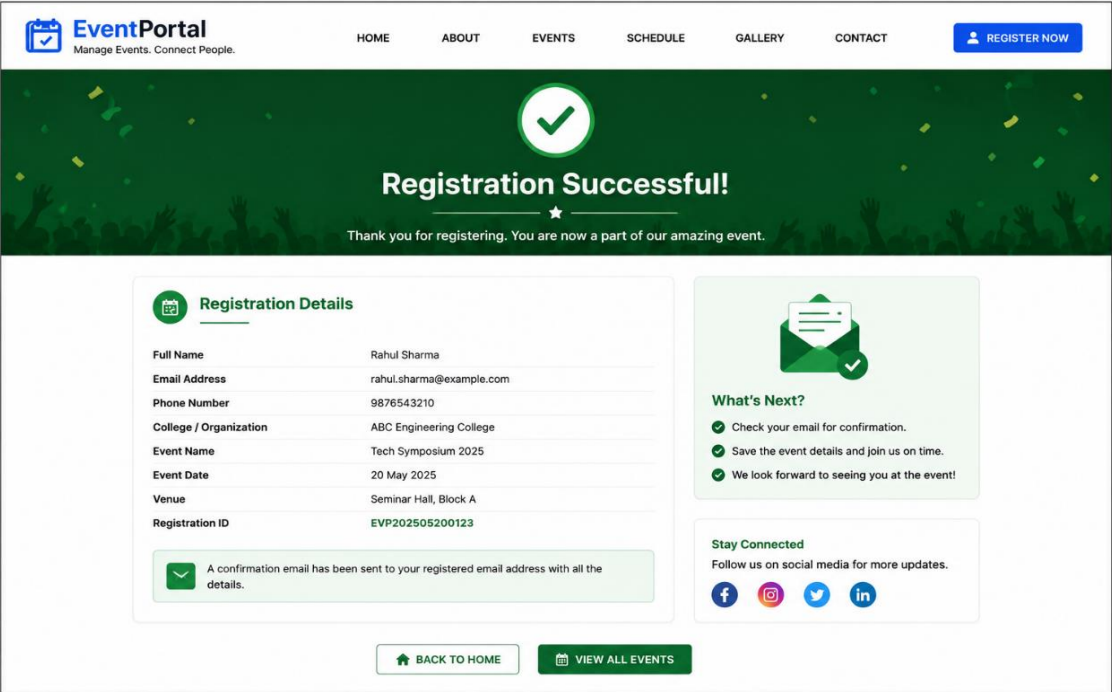
A platform for tech enthusiasts to learn, share and innovate with industry experts.

Important Notes

- Ensure all the details are correct before submission.
- You will receive a confirmation email after successful registration.
- Registration once submitted cannot be modified.
- For any queries, contact us through the Contact page.

Fig 5.4 Event Registration Page

Fig. 5.5 Registration Success Message



The screenshot shows the 'Registration Successful!' message page of the EventPortal. The header is identical to the registration page. The main heading is 'Registration Successful!' with a subtext 'Thank you for registering. You are now a part of our amazing event.' The page is divided into two main sections: 'Registration Details' and 'What's Next?'.

Registration Details

Full Name	Rahul Sharma
Email Address	rahul.sharma@example.com
Phone Number	9876543210
College / Organization	ABC Engineering College
Event Name	Tech Symposium 2025
Event Date	20 May 2025
Venue	Seminar Hall, Block A
Registration ID	EVP202505200123

☒ A confirmation email has been sent to your registered email address with all the details.

What's Next?

- ✓ Check your email for confirmation.
- ✓ Save the event details and join us on time.
- ✓ We look forward to seeing you at the event!

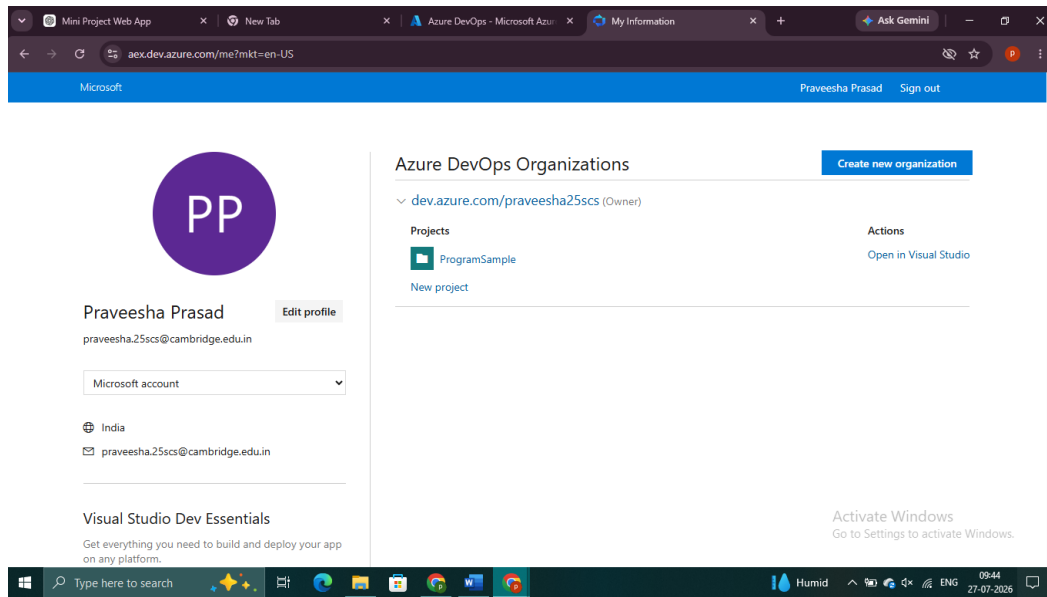
Stay Connected

Follow us on social media for more updates.

f i t in

BACK TO HOME VIEW ALL EVENTS

Fig 5.5 Registration Success Message



//

Fig. 5.6: Azure DevOps Pipeline

CHAPTER 7

CONCLUSION AND FUTURE SCOPE

The Cloud-Based Event Management Portal using Microsoft Azure and GitHub was successfully designed, developed, and deployed as a responsive web application that simplifies the process of managing events and participant registrations. The application provides users with an intuitive interface to browse upcoming events, view schedules, and register online while demonstrating the practical implementation of cloud computing technologies. The integration of GitHub with Azure DevOps Pipelines enables Continuous Integration and Continuous Deployment (CI/CD), ensuring that every code update is automatically built and deployed to Microsoft Azure App Service. The project achieves its objectives by providing a reliable, scalable, and user-friendly cloud-based event management solution.

The developed application demonstrates how cloud services can improve application accessibility, deployment efficiency, and maintainability. By utilizing Microsoft Azure for hosting and GitHub for version control, the project reduces manual deployment efforts and ensures that the latest version of the application is always available to users. The responsive design allows the portal to be accessed from various devices, making it suitable for educational institutions, organizations, and event organizers. Overall, the project provides a practical example of integrating modern web technologies with cloud computing services to develop an efficient event management system.

In the future, the Event Management Portal can be enhanced by integrating cloud databases such as Azure SQL Database to store participant and event information, implementing user authentication and administrator dashboards, and adding email or SMS notifications for event updates. Additional features such as online payment gateways, QR code-based event passes, analytics dashboards using Microsoft Power BI, AI-based event recommendations, and a dedicated mobile application can further improve the functionality and usability of the system. These enhancements would make the portal more secure, scalable, and suitable for real-world event management application.

